✅ Data Science Readiness for Production

Decision-First Checklist Template

Purpose: Determine whether a data science system is *ready to influence real decisions* — not whether it "works."

1️⃣ Decision Readiness

☐ The decision being supported is clearly stated in plain language
☐ The decision owner (human) is explicitly named
☐ The cost of being wrong is understood and documented
☐ Decision reversibility (or irreversibility) is clear
☐ This system is solving a *decision*, not "doing modeling for insight"

Red flags

- "We'll figure out how to use it later"
- Multiple stakeholders assume different decisions

2️⃣ Data & Assumption Readiness

☐ All major data sources are documented and defensible
☐ Known data limitations (delay, noise, bias) are acknowledged
☐ Label construction and trustworthiness are explicitly discussed
☐ Key assumptions are written down (not implied)
☐ There is agreement on what the data *does not* represent

Red flags

- "This is the best data we have"
- Labels treated as objective truth without context

3️⃣ Pipeline Coherence

☐ Cleaning strategy aligns with the decision's risk profile
☐ Feature representations reflect intended behavior signals
☐ Modeling approach (if used) satisfies decision constraints
☐ Evaluation metrics reflect real-world consequences

☐ Threshold strategies encode explicit policy choices
☐ No pipeline step contradicts earlier assumptions

Red flags

- Each stage optimized independently
- Metrics look good but outcomes feel wrong

4️⃣ Human Oversight & Delegation Boundaries

☐ Human review checkpoints are clearly defined
☐ AI roles (analyst, reviewer, implementer) are explicitly constrained
☐ AI is not allowed to make irreversible or policy decisions
☐ Humans know when *not* to trust the system
☐ Accountability for decisions is clearly human-owned

Red flags

- "The model decided"
- No clear intervention or override responsibility

5️⃣ Failure Awareness & Risk Acceptance

☐ Plausible failure scenarios are documented
☐ Silent failure modes are explicitly discussed
☐ The team knows how failure would *first* appear
☐ There is clarity on who absorbs cost when things go wrong
☐ Risks are consciously accepted, not ignored

Red flags

- "We'll catch it in monitoring"
- Surprise is treated as unavoidable

6️⃣ Organizational & Ethical Fit (Optional but Recommended)

☐ System behavior aligns with organizational values
☐ Uneven impact across groups has been considered
☐ The system can be explained retrospectively
☐ Deployment does not exceed the organization's maturity level
☐ There is confidence in defending this system externally if needed

Red flags

- "This is just an internal tool"
- "No one will notice if it's wrong"

🧭 Final Readiness Call

Overall assessment (choose one):

- ☐ ✅ Go
- ☐ ⚠️ Conditional Go
- ☐ ❌ No-Go

Justification (1–2 sentences):

Why is this decision system ready — or not ready — to ship?

🔑 Guiding Principle (Worth Remembering)

Production readiness is not about confidence. It's about clarity on what you are committing to.

How to Use This Template Effectively

- Run it **before** deployment discussions
- Use it to slow down "metrics-driven optimism"
- Treat a **No-Go** as a success outcome
- Revisit it when assumptions change

This checklist is designed to travel with you across roles, teams, and companies — it reflects **judgment**, not tooling.